CS-550 PA-4
Fatima Mariyam – A20527616
Performance Evaluation

Evaluation

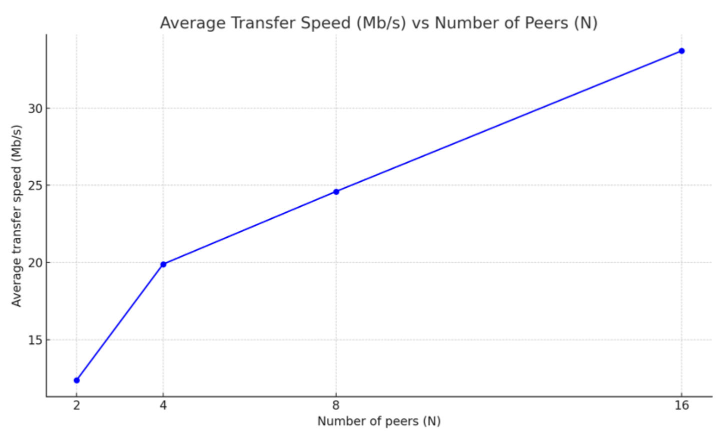
2. Measuring how the transfer speed changes when varying the number of nodes that hold the same requested file.
   a. Deploying 1 indexing server and a bunch of peers nodes.
   b. Varying the number of peer nodes (N) that holds 10 copies of the same files (N: 2, 4, 8, 16)
   c. Observe how the transfer speed changes when a peer node joins and requests to download these 10 files. (Each file is at least 1MB)
   d. Graph your results.

The experiment was performed for N=2,4,8 and 16 peers by querying file of size 84MB. (10 experiments)

Observation table:

Number of peers (N)	Average transfer speed (Mb/s)
2	12.4
4	19.9
8	24.6
16	33.7

Graph:



Observation:

The speed of transferring data increases as the number of peer nodes increases, and this can be attributed to several factors.

Firstly, parallelism comes into play when multiple peer nodes hold the same file. This allows the client to download different parts of the file simultaneously from multiple sources, leading to faster data transfer rates compared to downloading from a single source.

Secondly, when the number of peer nodes increases, the load on each individual peer decreases, reducing network congestion and contention for resources. This results in smoother and faster data transfers.

Improving Network Performance: By increasing the number of peer nodes, the client can make better use of the available network bandwidth. This can lead to higher overall throughput as the client can utilize multiple paths to download file segments.

Load Balancing: Distributing the download load across multiple peer nodes helps balance the load. Even if some peer nodes experience high demand, others can still serve the file, preventing bottlenecks and ensuring consistent transfer speeds.